

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Changes in Lipid Concentration” Equation

Proof. The equation is derived from the differential form of the continuity equation for mass:

$$\frac{\partial L}{\partial t} + \nabla \cdot \mathbf{J} + R = 0,$$

where L is the lipid concentration, $\mathbf{J}$ is the lipid flux, and R is the lipid degradation rate. The flux is described by the convection–diffusion equation

$$\mathbf{J} = -D_{\text{eff}} \nabla L + \mathbf{u}L,$$

where D_{eff} is the effective diffusion coefficient and $\mathbf{u}$ is the velocity field. Substituting this expression for $\mathbf{J}$ into the continuity equation gives

$$\begin{aligned} \frac{\partial L}{\partial t} &= -\nabla \cdot (-D_{\text{eff}} \nabla L + \mathbf{u}L) - R \\ &= \nabla \cdot (D_{\text{eff}} \nabla L) - \nabla \cdot (\mathbf{u}L) - R. \end{aligned}$$

For the lipid degradation model $R = 1/(kt + 1)$, the governing equation is:

$$\boxed{\frac{\partial L}{\partial t} = \nabla \cdot (D_{\text{eff}} \nabla L) - \nabla \cdot (\mathbf{u}L) - \frac{1}{kt + 1}}.$$